

Solution Architecture

Date	30 September 2026
Team ID	SWTID-2026-7960
Project Name	Import Data using Transform Maps (Spreadsheet) in ServiceNow
Maximum Marks	5 Marks

Solution Architecture Diagram:

1. Solution Architecture Mapping (Diagram Layout Fill):

- **Presentation / Client Layer:**

ServiceNow Service Portal / UI Navigator / Admin Dashboard (.csv/.xlsx upload interface)

- **API Gateway / Load Balancer:**

ServiceNow Platform Gateway & Node-level Routing

- **Auth Service:**

Role-Based Access Control (RBAC) & Platform Login Session Management

- **Core Logic Service:**

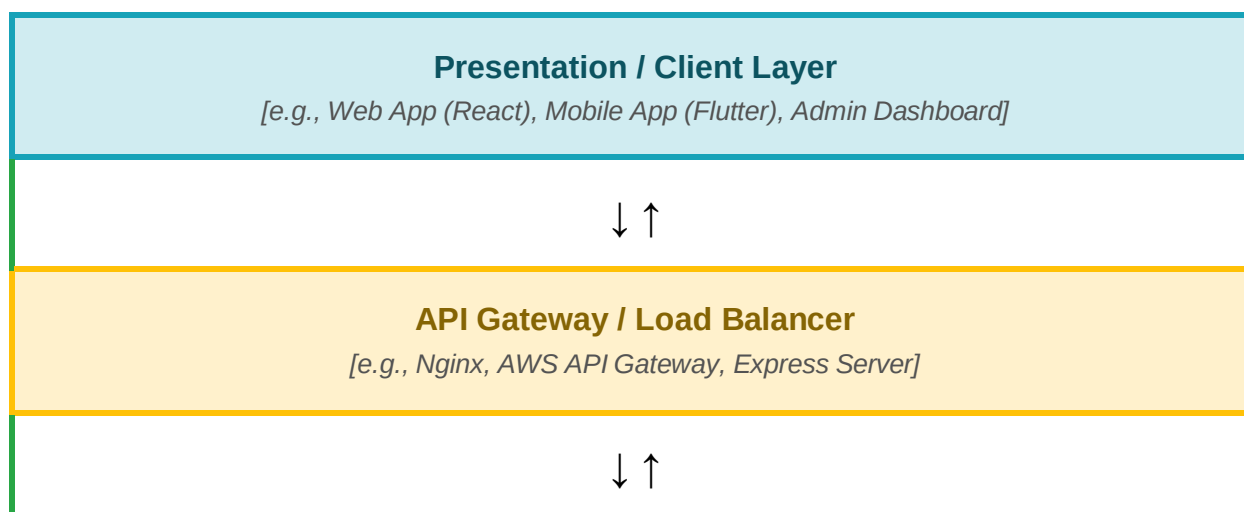
ServiceNow Import Sets, Transform Maps Engine, and Coalesce Matching Logic (u_employee_id)

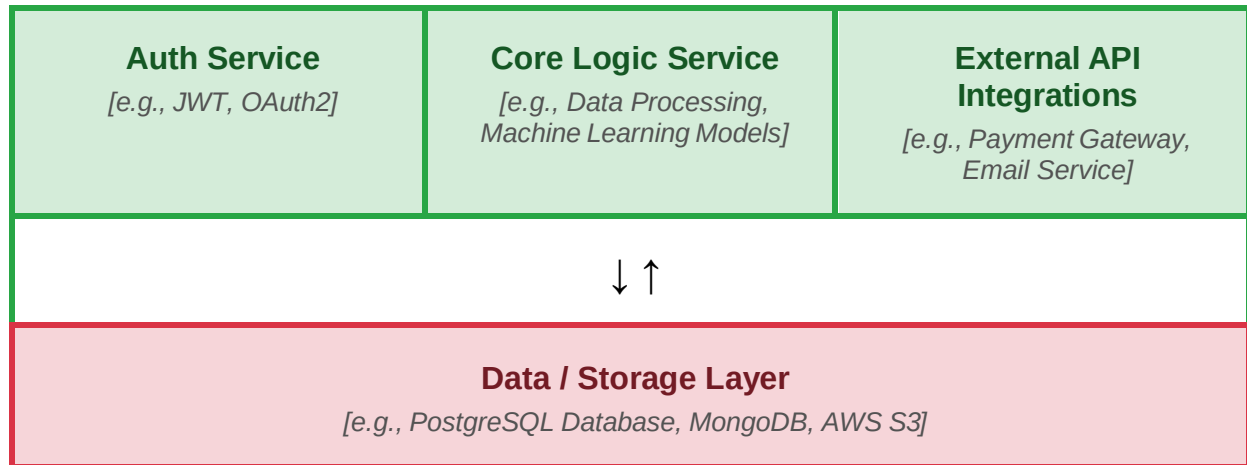
- **External API Integrations:**

Spreadsheet Parsers (.csv/.xlsx) & System File Import API

- **Data / Storage Layer:**

ServiceNow Relational Database (Staging Tables u_employee_import & Target Tables)





Component Description Tables

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for administrators to navigate modules, upload spreadsheets, and view data import dashboards.	ServiceNow Service Portal, HTML/CSS, UI Navigator
API Gateway	Manages incoming request routing, load distribution, and platform-level entry points across the ServiceNow instance.	ServiceNow Platform Gateway & Node Router
Microservices (Core Logic)	Handles staging data parsing, field mapping execution, and the Coalesce logic to prevent duplicate record insertion.	ServiceNow Transform Maps Engine, JavaScript, Server-Side Scripts
Database	Stores temporary raw records in staging tables and securely persists validated clean records in target tables.	ServiceNow Relational Database (MySQL-backed Staging & Target Tables)